

JAYASRI S

Curriculum Vitae

Bengaluru, India | jayasri.stat@gmail.com | LinkedIn

RESEARCH PROFILE

Early-career statistician and biostatistics researcher with experience in real-world evidence (RWE), oncology analytics, survival analysis, and causal inference using clinical and observational healthcare data. Experienced in applying statistical methodologies to patient registries, EHR, and OMOP-CDM datasets using R. Interested in methodological biostatistics research at the intersection of causal inference, translational medicine, and high-dimensional biomedical data.

RESEARCH INTERESTS

Biostatistics, Causal Inference, Survival Analysis, Real-World Evidence, Translational Medicine, Clinical Epidemiology, Longitudinal Data Analysis, High-Dimensional Healthcare Data

EDUCATION

Post Graduate Diploma in Biostatistics

Advanced Centre for Treatment, Research & Education in Cancer (ACTREC), Tata Memorial Centre, India
Completed: October 2024

Five-Year Integrated M.Sc. in Statistics

Pondicherry University, India
Completed: May 2023

RESEARCH & PROFESSIONAL EXPERIENCE

Data Scientist — RWE & Oncology Analytics

Molecular Connections Analytics Pvt. Ltd., Bengaluru, India
Dec 2024 – Present

- Conduct observational oncology analyses using real-world datasets to evaluate treatment patterns and comparative effectiveness using propensity score-based adjustment methods.
- Apply causal inference methodologies including propensity score matching (PSM) and inverse probability of treatment weighting (IPTW) on registry and claims datasets.
- Work with OMOP Common Data Model (CDM) frameworks for harmonisation and analysis of EHR and claims-based healthcare data.
- Develop reproducible analytical workflows in R for data preparation, quality control, and statistical reporting.
- Collaborate with clinicians, statisticians, and medical writers to translate clinical research questions into analytical methodologies and publication-oriented outputs.
- Developed Shiny-based tools for automated extraction of tables and figures from scientific PDFs to support evidence synthesis workflows and systematic reviews.

Biostatistician Trainee

ACTREC, Tata Memorial Centre (National Cancer Grid), Mumbai, India
Oct 2023 – Oct 2024

- Conducted survival analyses including Kaplan–Meier estimation, Cox proportional hazards modelling, and competing risks analysis on oncology research datasets.
- Contributed to Statistical Analysis Plans and protocol development for observational and interventional oncology studies.
- Generated statistical outputs, visualisations, and publication-oriented analyses using R.
- Provided statistical consultation to clinicians and postgraduate researchers regarding study design and analytical methodology.
- Selected for the CReDO Specialist Statistics Track under the National Cancer Grid (2024, 2025).

Biostatistics Intern — R&D Division

GlaxoSmithKline Pharmaceuticals Ltd., Bengaluru, India
Jan 2023 – Jul 2023

- Applied causal inference approaches and treatment effect estimation methods using oncology real-world

datasets.

- Reproduced clinical trial endpoints and generated TLFs for multiple myeloma studies using R and SAS.
- Gained exposure to CDISC SDTM/ADaM standards, clinical trial protocols, and statistical reporting workflows in a pharmaceutical R&D setting.

SELECTED PROJECTS

Clinical PDF Extraction Application

- Developed a Shiny-based application for automated extraction of tables and figures from scientific PDFs with Excel export and image-based table selection functionality.

Automated Antibiotic Treatment Analysis Workflow

- Designed reproducible analytical workflows integrating multiple healthcare datasets for CRE/CRPA treatment pattern analyses and reporting.

TEACHING & RESEARCH SUPPORT

- Delivered one-to-one and small-group training in R programming and survival analysis for clinical researchers and postgraduate students.
- Supported postgraduate students and research staff with data analysis, interpretation, and preparation of publication-ready results.
- Contributed to development of applied examples and analytical workflows for research methodology support.

TECHNICAL SKILLS

Statistical Programming: R, SAS, SQL, Python (basic)

Statistical Methods: Survival Analysis, Causal Inference, Longitudinal Data Analysis, Comparative Effectiveness Research, Observational Study Design

Healthcare Data Standards & Tools: OMOP-CDM, OHDSI Tools (ATLAS, CDMConnector), CDISC SDTM/ADaM, REDCap

Data Science & Reporting: R Shiny, R Markdown, ggplot2, dplyr, Tableau

CONFERENCES & ADVANCED TRAINING

- CReDO Specialist Statistics Track — Tata Memorial Centre & National Cancer Grid (2024, 2025)
- Survival Analysis Pre-Conference — Indian Society of Medical & Pediatric Oncology (2024)
- Symposium on Innovative and Applied Statistics (SIAS) — Novo Nordisk (2023)